

LaserSense

Low-Cost Laser-Based Long-Range Acoustic Sensing System

PROJECT DOCUMENTATION

Team name	Inovex
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Problem Statement	LaserSense: A Low-Cost Laser-Based Long-Range Acoustic Sensing System for Remote Situational Awareness
Event	HackMatrix 2026 – Round 2
Domain	Hardware / AI & ML / IoT / Emergency Response
Project Title	LaserSense – A Low-Cost Laser-Based Long-Range Acoustic Sensing System

Team Members

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Prajit R – Backend & System Integration Lead

Renu Shree S J – AI/ML & Signal Processing Lead

Viswashini SR – Frontend & Dashboard Lead

Links

GitHub Repository: https://github.com/viswashini22/laser_core

Live Deployed Link: <https://laser-core-5zxc.vercel.app/dashboard>

Demo Video: https://drive.google.com/file/d/1fqOP-JEoKHSkOIsRTuAnz3ePy2Y_xss1/view?usp=drivesdk

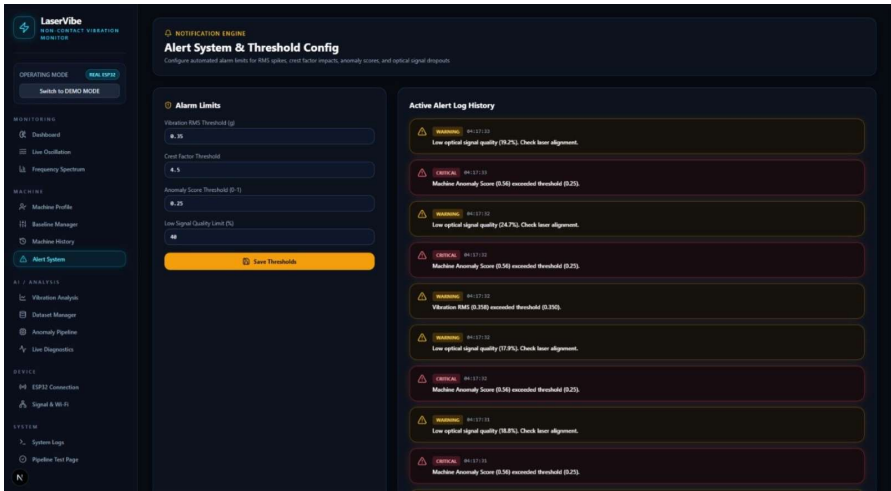
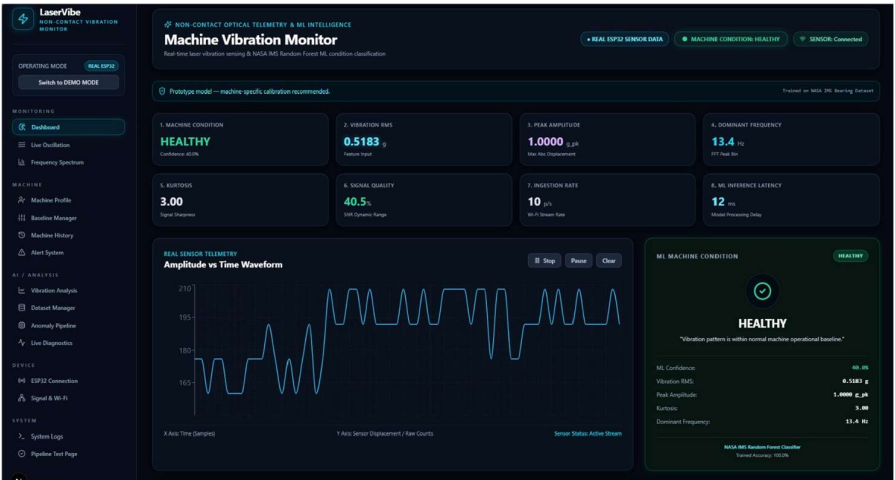
1. Project Overview

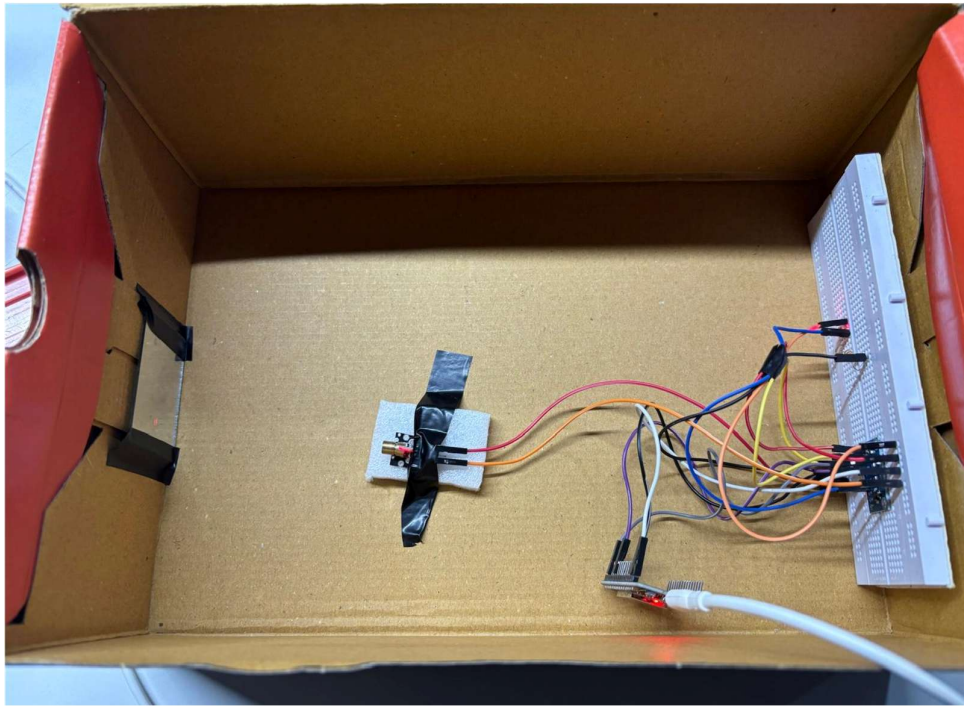
1.1 Summary

LaserSense is a low-cost, portable laser-based sensing system designed to detect vibration information from remote or difficult-to-access locations. A 650 nm laser illuminates a vibrating surface, and changes in the reflected laser signal are captured using a photodiode. The acquired signal is digitized and processed to extract meaningful vibration patterns, with Python-based signal analysis and AI-assisted processing supporting acoustic or event detection.

1.2 Platform Preview

The software platform provides a live monitoring dashboard for presenting processed vibration information and detected events. The dashboard is intended to make the output understandable to operators by combining signal visualization, analysis results and alert/status information.





1.3 Working Flow

Machine or acoustic activity creates micro-vibrations on a target surface.

A 650 nm laser illuminates the vibrating surface from a distance.

The reflected laser carries information related to the surface vibration.

A photodiode detects variations in the reflected optical signal.

An ADC and ESP32 acquire the digital signal for transmission.

Python-based processing performs noise reduction, FFT and signal analysis.

AI/ML processing supports classification or meaningful event detection.

The live dashboard presents the processed output and alerts.

1.4 Objective

The main objective is to demonstrate that a low-cost optical sensing setup can provide useful remote vibration information without requiring a conventional microphone to be physically placed at the target.

2. Problem Being Solved

Remote acoustic and vibration monitoring becomes difficult when the sensing point is hazardous, inaccessible or too far away for conventional contact or proximity-based sensing. Commercial remote laser sensing equipment can also be expensive, creating a barrier to wider deployment. In addition, raw vibration signals may be weak or affected by environmental noise, so useful information requires appropriate signal processing and intelligent interpretation.

2.1 Key Causes

Physical access requirement – conventional sensors generally need to be positioned close to the source or target.

Weak and noisy remote signals – long-range measurements can contain environmental interference and require reliable processing.

Limited signal interpretation – raw vibration data needs analysis to identify meaningful patterns or events.

2.2 Proposed Solution

LaserSense addresses these challenges through a non-contact optical sensing approach. The system uses a 650 nm laser and photodiode to capture remote surface-vibration information, while an ESP32-based acquisition unit transfers the data for software processing. Python, NumPy and SciPy are used for signal analysis such as FFT and related feature calculations, while the trained ML workflow supports vibration/event classification. The processed information is presented through a real-time dashboard.

2.3 USP – Unique Selling Point

LaserSense combines low-cost laser-based non-contact sensing, embedded data acquisition, signal processing and AI-assisted event detection into a portable monitoring workflow.

2.4 Who Benefits?

Disaster response and emergency rescue teams

Industrial safety and inspection teams

Operators responsible for critical infrastructure

Teams that need situational information without physically approaching the sensing location

2.5 Expected Value

The system aims to enable remote monitoring of industrial machine vibrations, helping detect abnormal vibration patterns, identify potential machine faults at an early stage, reduce the need for direct physical inspection, and provide a low-cost platform for vibration analysis and predictive maintenance.

3. Key Features

- Non-contact laser-based vibration sensing 650 nm laser illumination
- Photodiode-based optical signal detection
- ESP32-based data acquisition and communication
- ADC-based analog measurement
- Python-based signal processing
- Noise reduction and FFT analysis
- Feature extraction for vibration analysis
- Real-time dashboard with vibration visualization
- Alert/status presentation for detected events

4. Technical Architecture

The implementation combines hardware sensing, embedded acquisition, backend services, signal processing, machine learning and a web dashboard. The dashboard architecture uses React.js/Next.js with Tailwind CSS and Recharts for the user interface, while Python with FastAPI provides the backend API.

4.1 Technology Stack

Frontend	React.js / Next.js
UI Styling	Tailwind CSS
Charts	Recharts / Chart.js
Backend API	Python + FastAPI
Communication	REST API / WebSocket
Signal Processing	NumPy + SciPy
AI/ML	Python + Scikit-learn

ML Model	Random Forest
Model Storage	Joblib
Database	Firebase / PostgreSQL
Hardware	ESP32
Laser Sensor	Laser module + photodiode/LDR
Deployment	Vercel + Render/Railway
Dataset	NASA IMS Bearing Dataset – initial ML training

4.2 Data Flow

Target vibration → Laser illumination → Reflected laser → Photodiode detection → ADC → ESP32 acquisition → USB/Wi-Fi transmission → Python signal processing → AI/ML classification → Dashboard visualization and alerts.

5. Implementation Status

5.1 Already Built

Dataset-based AI prediction/classification model

Signal analysis workflow

Dashboard for monitoring and visualization

Backend/API structure for serving processed results

Initial integration of the software components

5.2 Planned Next Steps

Hardware integration with the sensing and acquisition pipeline

Real-time data integration from ESP32

Field testing under controlled conditions

Further model validation and refinement

6. Future Scope

Improve robustness against environmental noise and changing operating conditions.

Extend testing to different target surfaces, distances and vibration conditions.

Integrate the physical sensing unit more tightly with the deployed dashboard.

Expand the event-classification dataset and improve model generalization.

Evaluate the system for industrial safety, emergency response and critical infrastructure scenarios.

Develop a more compact and field-ready sensing unit for practical deployment.

7. Why Our Team Can Execute the Solution

Our team combines IoT and ESP32 expertise, backend and system integration, AI/ML and signal processing, and frontend dashboard development to build and connect the LaserSense pipeline end-to-end.

8. Conclusion

LaserSense presents a practical approach to remote vibration sensing by combining low-cost optical sensing with embedded acquisition, digital signal processing, machine learning and real-time visualization. The current software prototype establishes the prediction and dashboard workflow, while future hardware integration and field validation will move the system toward a complete real-world sensing platform.